147 Major St — HVAC clash mark-up

HVAC Designs LO# 113963 (HVA 26 167838) vs permit 26 167838 BLD 00 · PalmView BIM review · Oct 22, 2026

Two systems: F-1 basement (basement + main floor from below) · F-2 2nd-floor mech room (2nd + 3rd floors from the 3rd-floor joist space).

Questions / open items

Q1 **2nd-floor ceilings for the F-2 trunks**

Rear (lower) level, 10'-4½" ceiling: drop the hall / bath / laundry / W.I.C. / bedroom 2 ceilings flat to ~9'-6" to hide C, B, Y, 6R and branches 25-28, 33/34 (or box them at 9'-8"). Front (upper) level, 9'-0½" ceiling: trunk A along the south wall and return Z in the family room need 14" bulkheads at 8'-4", or a reroute. Owner: PalmView / client. Status: RESOLVED.

Q2 **Bedroom 2 ceiling: joist direction and trunk B on LVL1**

Confirm the cantilever TJIs over bedroom 2 run front-to-back (balcony). Trunk B sits on the south LVL1; 5 branches cross 26 cantilever joists. Owner: Kott / Rona + engineer. Status: SITE.

Q3 **Rear branches 8, 10, 23 through the old rear foundation**

Core three holes in the brick, or route all rear branches through the existing 30" opening. Owner: HVAC sub. Status: OPEN.

Q4 **Supply 9 and return 8R sit on old foundation walls**

Both stacks land on top of existing masonry (powder E wall, pantry/storage double wall). Relocate. Owner: HVAC Designs. Status: OPEN.

Q5 **Return 3R on the ensuite vanity / mirror wall**

High-wall grille + 7" stack behind the 72" vanity, mirrors, sconces and feature tile. Owner: HVAC Designs / designer. Status: OPEN.

Q6 **7" return stacks in 2x4 walls (1R–6R, 9R) and 6" supply stacks (9, 24)**

All stick out 3–4" past the studs. 3-1/4x14 stacks, 2x6 walls, or chases. Owner: HVAC sub / framer. Status: OPEN.

Q7 **Return 4R stack vs the LVL5 flush beam**

Grille is on the closet wall right under LVL5; the stack cannot pass the beam. Owner: HVAC Designs / engineer. Status: RESOLVED.

Q8 **Kitchen sizes: range, fridge, run length and aisle**

Permit shows a 30" range and 34" fridge; Miele range and fridge are 36". Cabinet set 186.5" vs 184.6" wall. Aisle to the stair is 34.9". Owner: Cabinet shop / PalmView. Status: OPEN.

Q9 **Kitchen hood exhaust not designed**

Elica Trento insert on the south wing wall under the 2nd-floor overhang. No duct or termination drawn. Owner: HVAC Designs. Status: OPEN.

Q10 **F-1 new or existing; hot water tank vs plenum**

F-1 not tagged existing and sits where the permit shows the existing H.E. furnace. Takeoff runs over the tank vent. Owner: PalmView. Status: OPEN.

Q11 **Stair direction 2nd → 3rd, and the size of the 2nd-floor step**

Plans A8/A9: new stair stacked over the existing one, rising front-to-rear. Section A-A draws it rising rear-to-front. Step: section A-A framing shows 16" (2R at 8"); the rear and south elevations show 1'-9" (10'-11" vs 9'-2"). Model uses 16". Owner: PalmView / designer. Status: OPEN.

Q12 **Paperwork and missing items**

No heat loss/gain, EEDS or Schedule 1 in the package; HRV models not stated; 1 basement return in the schedule not drawn; exhaust routes not drawn; 2R drawn where the permit shows no wall. Owner: HVAC Designs. Status: OPEN.

Q13 **Front roof space over the primary bedroom**

Trunk A (first 6.7 ft) and branches 35/36 are under the existing front roof. If that space is outside the insulation, ducts need R-12 + Class A sealing (permit note). Owner: PalmView. Status: OPEN.

Resolved with Raff

Oct 22 - Shifts made with the drop (HVAC sub to follow)

Hung under the joists, three runs moved off wall top plates: trunk C 2" east (was 1.4" into the notch wall), return Y 5" east (was over the bath east wall), 6R run and stack 5" east (was over the W.I.C./mech wall). Still to place on site: the C-to-A tie (10x8 at the step line) lands between the family-room rear wall and the bath north wall, which are only 7.8" apart - box it on the family-room side.

Oct 22 - 2nd floor is split-level

Front (existing house, y < 28 ft) is the upper level at 113.85; the kitchen-wing / addition part steps down. Measured 16" from the section A-A framing; UP 2R at the stair landing; flush LVL5 on the step line. Ground stair 17R lands on the lower level. Model updated.

Oct 22 - Side-yard notch is full height - no overhang

The south wall jogs in to x 4.2-5.0 ft from y 28 to 44 ft at every level, basement to third floor. Removed the 2nd/3rd-floor cantilever over the side yard; bath and ensuite windows moved onto the notch wall. HVAC Designs backgrounds already show it correctly.

Not connected (no route drawn)

- Bath fan 5"Ø - basement bath
- Bath fan 5"Ø - powder (Panasonic 80-110 cfm ML-036)
- Bath fan 5"Ø - 2nd floor bath
- Exhaust 5"Ø - laundry
- Bath fan 5"Ø - ensuite (Panasonic 80-110 cfm ML-068)
- Kitchen hood - Elica Trento ETR134S1 insert
- Basement return grille (schedule says BAS R/A = 1)

Legend

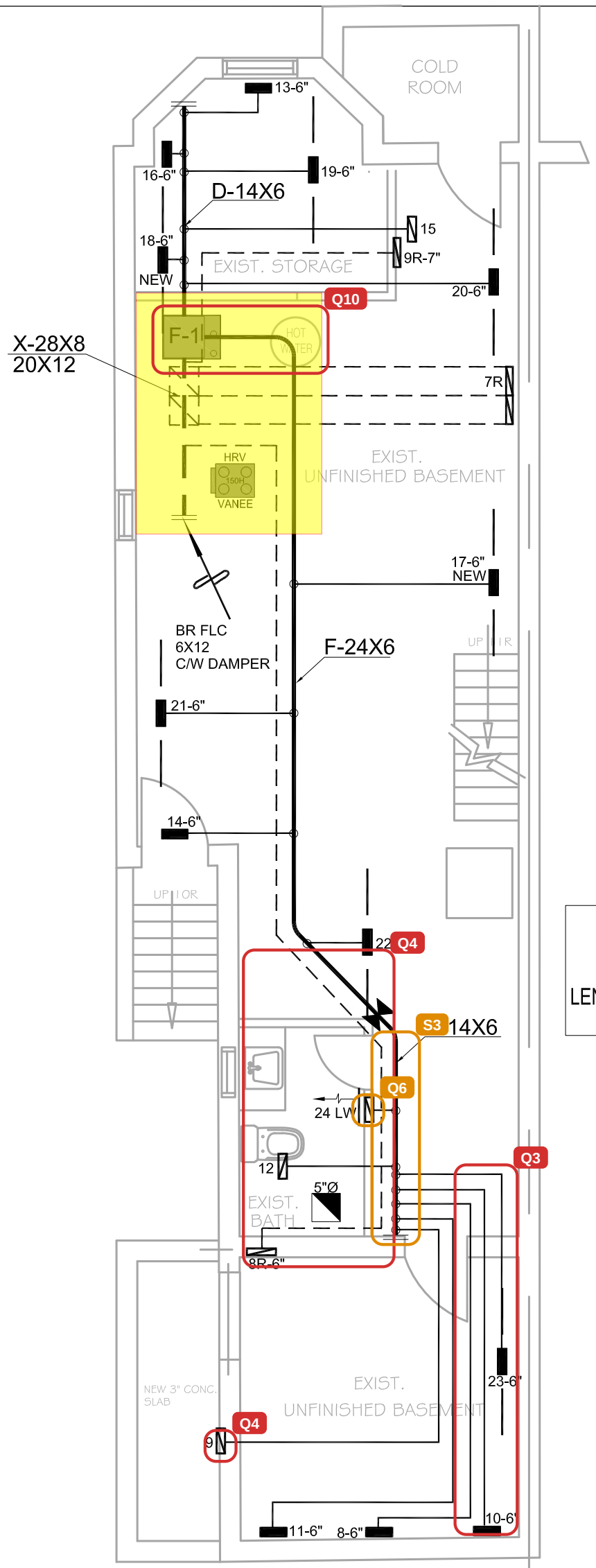
Hard clash / decision needed

Soft clash, site check, or stack too deep for its wall

Drawing conflict between sheets

SUPPLEMENTAL EXHAUST FAN (MINIMUM 100 CFM) DISCHARGING DIRECTLY TO THE EXTERIOR WILL BE PROVIDED IN THE KITCHEN. EXHAUST DUCTS SERVING THE KITCHEN SHALL NOT BE SMALLER THAN RECOMMENDED BY THE MANUFACTURER OF THE FANS AND MUST AT LEAST MEET O.B.C. STATED MINIMUMS.

ENSURE A/C COIL AND OUTDOOR UNIT MEET AHRI CERTIFICATION AS REQUIRED. CONTRACTOR TO PROVIDE AHRI CERTIFICATE PRIOR TO INSTALLATION FOR REVIEW.



PROPOSED BASEMENT PLAN

SCALE: 3/16"=1'-0"

PALMVIEW CLASH REVIEW · OCT 22, 2026

Q3 Rear branches 8, 10, 23 through the old rear foundation

Q4 Supply 9 and return 8R sit on old foundation walls

Q6 7" return stacks in 2x4 walls (1R–6R, 9R) and 6" supply stacks

Q10 F-1 new or existing; hot water tank vs plenum

S3 Trunk E ends ½" off the old rear wall

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TORONTO Building

PERMIT REVIEWED FOR COMPLIANCE WITH THE ONTARIO BUILDING CODE

26 167838 HVA 00

ZONING		
O.B.C.	Wu, Tony	02/Sep/2026
FIRE SERVICES		

O.B.C. (S)

FURNACE 1
LENNOX ML196UH070XE36B
63,900 BTU/H OUTPUT
LENNOX ML14KP1 - 1.5 TON HEAT PUMP
928 DESIGN CFM

	S/A	R/A	FANS
3RD	8	3	1
2ND	6	3	2
1ST	9	3	2
BAS	8	1	0

HVAC LEGEND

SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	SUPPLY AIR GRILLE		6" SUPPLY AIR BOOT ABOVE		14"x8" RETURN AIR GRILLE		RETURN AIR STACK ABOVE
	SUPPLY AIR GRILLE 6" BOOT		SUPPLY AIR STACK FROM 2nd FLOOR		30"x8" RETURN AIR GRILLE		RETURN AIR STACK 2nd FLOOR
	SUPPLY AIR BOOT ABOVE		6" SUPPLY AIR STACK 2nd FLOOR		FRA- FLOOR RETURN AIR GRILLE		REDUCER

No.	Description	Date
3.		
2.		
1.	REVISED AS PER COMMENTS	AUG/2026

REVISIONS

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I Michael O'Rourke have reviewed and take responsibility for the design work and am qualified under division C,3.2.5 of the building code.

Mlchael O'Rourke BCIN # 19669
HVAC Designs Ltd.

CUSTOM

Client
PALMVIEW HOMES

Project Name
147 MAJOR STREET
TORONTO, ONTARIO

CUSTOM 2628 sqft

HVAC DESIGNS LTD.
375 Finley Ave. Suite 202 - Ajax, Ontario
L1S 2E2 Tel. 905.619.2300 - 905.420.5300 Fax 905.619.2375
Email: info@hvacdesigns.ca
Web: www.hvacdesigns.ca
Specializing in Residential Mechanical Design Services

Installation to comply with the latest Ontario Building Code. All supply branch outlets shall be equipped with a manual balancing damper. Ductwork which passes through the garage or unheated spaces shall be adequately insulated and be gas-proofed.

Sheet Title
BASEMENT HEATING LAYOUT

Date JULY/2026

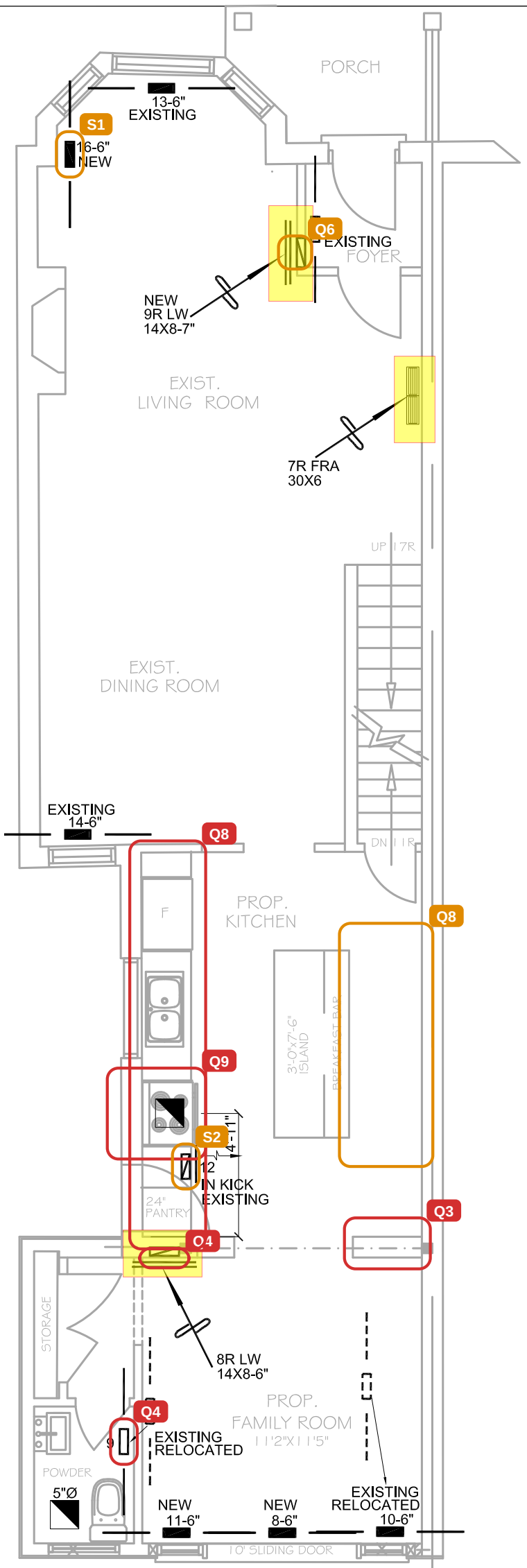
Scale 3/16" = 1'-0"

BCIN# 19669

LO# 113963

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TORONTO Building

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ZONING		
O.B.C.	Wu, Tony	02/Sep/2026
FIRE SERVICES		

O.B.C. (S)

PALMVIEW CLASH REVIEW · OCT 22, 2026

- Q3 Rear branches 8, 10, 23 through the old rear foundation
- Q4 Supply 9 and return 8R sit on old foundation walls
- Q6 7" return stacks in 2x4 walls (1R–6R, 9R) and 6" supply stacks
- Q8 Kitchen sizes: range, fridge, run length and aisle
- Q9 Kitchen hood exhaust not designed
- S1 Supply 16 boot tight to the bay corner
- S2 Supply 12 kick register under the pantry

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PROPOSED GROUND FLOOR PLAN

SCALE: 3/16" = 1'-0"

HVAC LEGEND

SUPPLY AIR GRILLE		6" SUPPLY AIR BOOT ABOVE		24"x8" RETURN AIR GRILLE	RETURN AIR STACK ABOVE
SUPPLY AIR GRILLE 6" BOOT	⊘	SUPPLY AIR STACK FROM 2nd FLOOR		30"x8" RETURN AIR GRILLE	RETURN AIR STACK 2nd FLOOR
SUPPLY AIR BOOT ABOVE		6" SUPPLY AIR STACK 2nd FLOOR		FRA- FLOOR RETURN AIR GRILLE	REDUCER

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Michael O'Rourke BCIN # 19669
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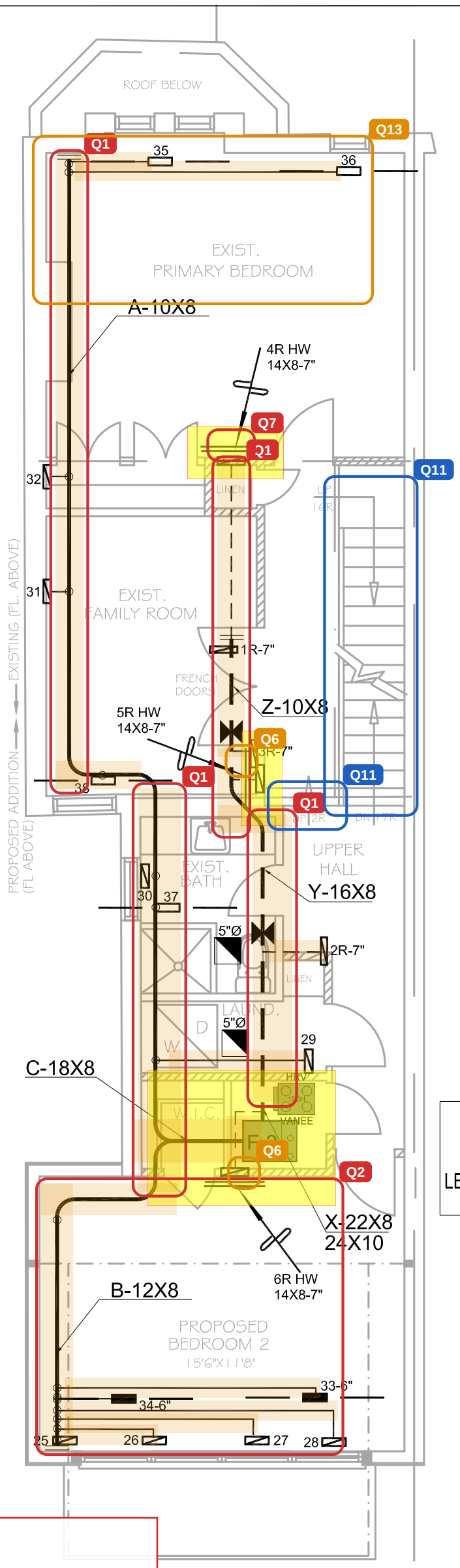
Installation to comply with the latest Ontario Building Code. All supply branch outlets shall be equipped with a manual balancing damper. Ductwork which passes through the garage or unheated spaces shall be adequately insulated and be gas-proofed.

Sheet Title
FIRST FLOOR HEATING LAYOUT

Date JULY/2026
Scale 3/16" = 1'-0"
BCIN# 19669
LO# 113963

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FURNACE 2
LENNOX ML196UH030XE36B
29,000 BTU/H OUTPUT
LENNOX ML14KP1 - 2.0 TON HEAT PUMP
735 DESIGN CFM

PALMVIEW CLASH REVIEW · OCT 22, 2026

Q1 2nd-floor ceilings for the F-2 trunks

Q2 Bedroom 2 ceiling: joist direction and trunk B on LVL1

Q6 7" return stacks in 2x4 walls (1R–6R, 9R) and 6" supply stacks

Q7 Return 4R stack vs the LVL5 flush beam

Q11 Stair direction 2nd → 3rd, and the size of the 2nd-floor step

Q13 Front roof space over the primary bedroom

PROPOSED SECOND FLOOR PLAN

SCALE: 3/16" = 1'-0"

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O.B.C.	Wu, Tony	02/Sep/2026
FIRE SERVICES		

O.B.C. (S)

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HVAC LEGEND

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	SUPPLY AIR GRILLE 6" BOOT		SUPPLY AIR STACK FROM 2nd FLOOR		30"x8" RETURN AIR GRILLE		RETURN AIR STACK 2nd FLOOR
	SUPPLY AIR BOOT ABOVE		6" SUPPLY AIR STACK 2nd FLOOR		FRA- FLOOR RETURN AIR GRILLE		REDUCER

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Michael O'Rourke BCIN # 19669
HVAC Designs Ltd.

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Sheet Title
SECOND FLOOR HEATING LAYOUT

Date JULY/2026
Scale 3/16" = 1'-0"
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LO# 113963

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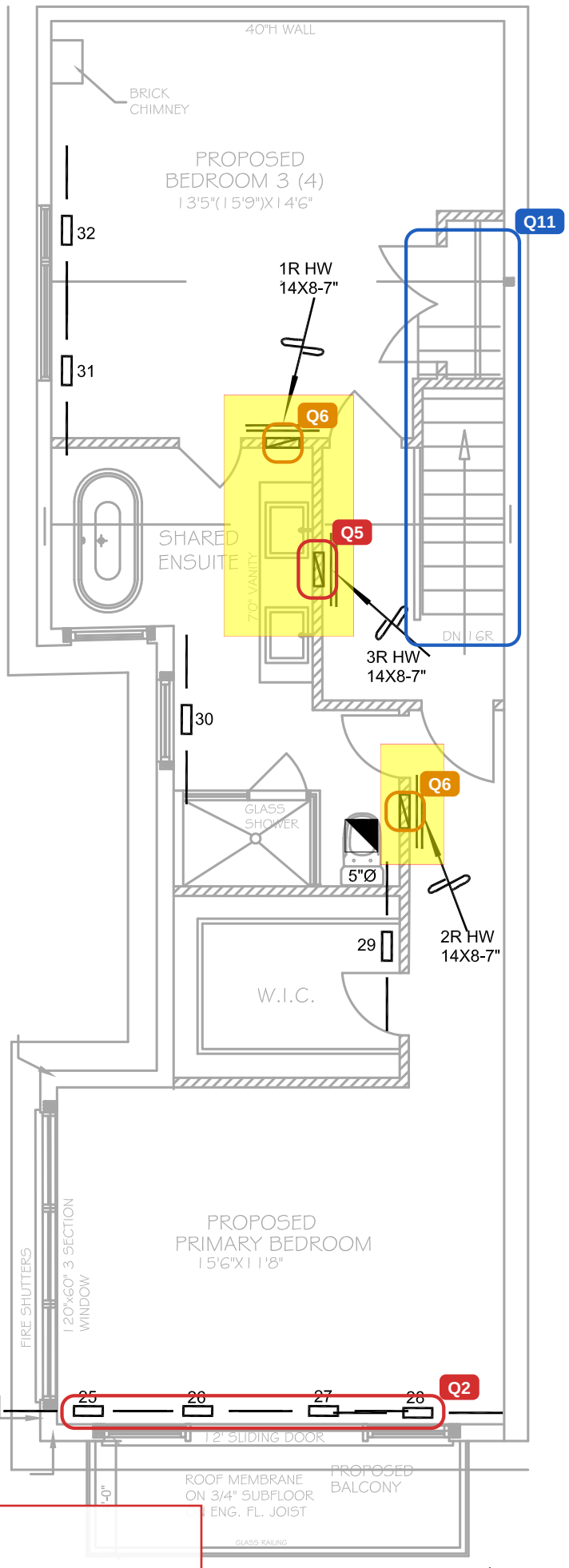
TORONTO Building

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FIRE SERVICES		

O.B.C. (S)



PROPOSED THIRD FLOOR PLAN

SCALE: 3/16" = 1'-0"

PALMVIEW CLASH REVIEW · OCT 22, 2026

Q2 Bedroom 2 ceiling: joist direction and trunk B on LVL1

Q5 Return 3R on the ensuite vanity / mirror wall

Q6 7" return stacks in 2x4 walls (1R–6R, 9R) and 6" supply stacks

Q11 Stair direction 2nd → 3rd, and the size of the 2nd-floor step

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HVAC LEGEND

SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	SUPPLY AIR GRILLE		6" SUPPLY AIR BOOT ABOVE		14"x8" RETURN AIR GRILLE		RETURN AIR STACK ABOVE
	SUPPLY AIR GRILLE 6" BOOT		SUPPLY AIR STACK FROM 2nd FLOOR		30"x8" RETURN AIR GRILLE		RETURN AIR STACK 2nd FLOOR
	SUPPLY AIR BOOT ABOVE		6" SUPPLY AIR STACK 2nd FLOOR		FRA- FLOOR RETURN AIR GRILLE		REDUCER

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Sheet Title
THIRD FLOOR HEATING LAYOUT

Date JULY/2026

Scale 3/16" = 1'-0"

BCIN# 19669

LO# 113963